Hypothesis Testing on normality check

Although QQ-plot can provide a quick scan on the validity of the normality assumption, it may not able to draw an accurate conclusion since it is a kind of graphical method. In order to get a more precise prediction, we will examine this using hypothesis testing.

There are a lot of different test available for this purpose. In this section, we will explore the most intuitive test so called *Kolmogorov-Smirnov test*.

Principle of Kolmogorov-Smirnov test (KS-test)

This test aims to test whether the sample data comes from a certain probability distribution (denoted by $F_X(x)$).

Under this test, the two hypothesis will be set as follows:

H_0: $The\ sample\ data\ is\ drawn\ from\ the\ probability\ distribution\ F_X(x)$

H_1: $The\ sample\ data\ is\ NOT\ drawn\ from\ the\ probability\ distribution\ F_X(x)$

(*In our case, this distribution $F_X(x)$ is the normal distribution)

What is the test statistic?

Given the sample data $\{X_1, X_2, \dots, X_n\}$, we let $F_n(x)$ be the empirical distribution of Z obtained from this sample data.

When n is sufficiently large, $F_n(x)$ can roughly approximate the true distribution. So one can check the normality by comparing the empirical distribution with the normal distribution (denoted by $F_X(x)$) and see whether there is any significant difference.

Therefore, we consider the following statistics:

$$D_n = \sup_{x \in (-\infty, \infty)} |F_n(x) - F(x)| \geq 0.$$

If X_i is normally distributed, then D_n should be very close to 0 especially when $n \to \infty$. Thus, we deduce that

$$\left\{ \begin{matrix} Reject\ H_0\ \ and \\ Accept\ H_1 \end{matrix} \right\} \Leftrightarrow D_n > D^*,$$

where D^* is a critical threshold.

Assuming H_0 is true, one can show that when $n \to \infty$, the test statistic $\sqrt{n}D_n$ has the following approximated distribution:

$$P\left(\underbrace{\sqrt{n}D_n}_{denoted\ by\ K} \leq x\right) \approx \frac{\sqrt{2\pi}}{x}\sum_{k=1}^{\infty} e^{-\frac{(2k-1)^2\pi^2}{8x^2}}.$$

Given the significant level α, we choose the critical value K_α such that

$$P(K > K_\alpha) = \alpha.$$

When the sample size is large ($n \geq 50$) and the $F(x)$ is the c.d.f. of normal distribution, the critical value of K_α is given as follows:

α	0.001	0.01	0.05	0.1
K_α	1.94947	1.62762	1.3581	1.22385

In R, one can perform the test by following command (for checking normality):

ks.test(x, ”pnorm”, mean=??, sd=??)

(*Remark: “pnorm” can be replaced by other sample data y or other distributions).